

## Compositional Learning of Dynamical System Models Using PHNNs Roadmap

1. Hamiltonian Systems Theory (HST) → Hamiltonian Neural Networks (HNN): Closed Systems
2. Port (PHST) framework → Port Hamiltonian Neural Network (PHNN): Open Physical Systems
3. Converting from Dirac PHST to Monotonicity PHST → incremental (iPHST): General Systems
4. *incremental PHST* → *IPHNN (incremental Port Hamiltonian Neural Networks)*

### **Why HNN (Ref: [Hamiltonian Neural Networks](#))? General neural networks drifts from true dynamics**

The first problem is the notion of discrete “time steps” that connect neighboring states. Since time is actually continuous, a better approach would be to express dynamics as a set of differential equations and then integrate them from an initial state at  $t_0$  to a final state at  $t_1$ .

Second problem with existing methods is that they tend not to learn exact conservation laws. This often causes them to drift away from the true dynamics of the system as small errors accumulate.

### **HNN works for well-defined physical systems. But what about nonphysical dynamical systems?**

### **PHNN (Ref [Port Hamiltonian Neural Networks](#))? Dynamical systems with energy dissipation**

Hamiltonian or Lagrangian neural networks predict trajectories of physical systems and they generally tackle autonomous systems that depend implicitly on time or with a known control signal.

Real world dynamical systems are non-autonomous, driven by time-dependent forces and experience energy dissipation. Hence, this challenge of learning from such non-autonomous systems is addressed by embedding the port-Hamiltonian framework that can capture energy dissipation and time-dependent control forces.

Port-Hamiltonian neural networks can efficiently learn the dynamics of nonlinear physical systems of practical interest and accurately recover the underlying stationary Hamiltonian, time-dependent force, and dissipative coefficient.

## **Ideas**

### **1) Incremental Port HNN (IPHNN) (Ref: [Incrementally port-Hamiltonian systems](#) IEEE 2013)**

- Develop or design NN based on incremental PHST framework using monotone relation.
- By replacing the composition of a Dirac structure and an energy dissipating relation by a maximal monotone relation.
- Composition of two incrementally port-Hamiltonian systems is again an iPHS based on the fact that the composition of two maximal monotone relations is again maximal monotone.
- Gradients of convex functions are cyclically monotone.

### **2) Port HST and Monotonicity (Ref: [Port-Hamiltonian systems and monotonicity](#) 2022)**

- Understand how to extend incremental (PHS) to maximal cyclically monotone relations.
- Understand compositions of maximal monotone and maximal cyclically monotone.

### **3) Port Hamiltonian Gradient Flows (Ref: [Port-Hamiltonian Gradient Flows](#) ICLR 2020)**

- A general framework for continuous-time gradient descent (gradient flow).
- Extension of Hamiltonian gradient flows using a dynamical system perspective.
- Using [Port-Hamiltonian Approach to Neural Network Training](#) (IEEE CDC 2019)